

EDUCATION

University of California, Los Angeles
Ph.D. Candidate in Computer Science.
Advisor: Quanquan Gu

Los Angeles, CA, USA
09/2023 - Current

University of California, Los Angeles
M.Sc. in Computer Science

Los Angeles, CA, USA
09/2023 - 12/2025

Tsinghua University
B.E. in Computer Science and Technology

Beijing, China
08/2019 - 07/2023

RESEARCH INTERESTS

Machine Learning, Reinforcement Learning, Information Theory
Reinforcement Learning and Efficient Methods for Large Language Models

RESEARCH AND WORKING EXPERIENCE

Knowledge Engineering Group, Tsinghua University
Research Assistant, Advisor: Prof. Jie Tang

07/2021 - 01/2022

- Developed P-tuning V2 method (a prefix-tuning-like method) that is comparable with full-finetune universally across model scales and tasks.
- Investigated the performance of P-tuning on neural text retriever.

Knowledge Engineering Group, Tsinghua University
Research Assistant, Advisor: Prof. Juanzi Li

03/2023 - 06/2023

- Developed efficient in-context method for LLM to solve open information retrieval task.
- Investigated LLM's ability in understanding video when visual tools are available.

ByteDance Inc.

06/2024 - 09/2025

Research Scientist Intern, Mentor: Renjie Zheng

- Designed actor-critic reinforcement learning algorithm for LLM post-training
- Developed direct-preference-learning styled offline RL algorithm to enhance reasoning ability of LLMs.
- Applied reinforcement learning methods for training memory agent.

RESEARCH PUBLICATIONS

- [1] Qiwei Di*, **Kaixuan Ji***, Xuheng Li*, Heyang Zhao, and Quanquan Gu. Best-of-majority: Minimax-optimal strategy for pass@k inference scaling. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- [2] Qingyue Zhao*, **Kaixuan Ji***, Heyang Zhao*, Tong Zhang, and Quanquan Gu. Towards a sharp analysis of offline policy learning for f -divergence-regularized contextual bandits. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- [3] **Kaixuan Ji***, Jiafan He*, and Quanquan Gu. Reinforcement learning from human feedback with active queries. *Transactions on Machine Learning Research (TMLR)*, 2025. **Featured Certification**.
- [4] Huizhuo Yuan*, Zixiang Chen*, **Kaixuan Ji***, and Quanquan Gu. Self-play fine-tuning of diffusion models for text-to-image generation. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024.

- [5] Yue Wu*, Zhiqing Sun*, Huizhuo Yuan*, **Kaixuan Ji**, Yiming Yang, and Quanquan Gu. Self-play preference optimization for language model alignment. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
- [6] Zixiang Chen*, Yihe Deng*, Huizhuo Yuan*, **Kaixuan Ji**, and Quanquan Gu. Self-play fine-tuning converts weak language models to strong language models. In *Forty-first International Conference on Machine Learning (ICML)*, 2024.
- [7] **Kaixuan Ji***, Qingyue Zhao*, Jiafan He, Weitong Zhang, and Quanquan Gu. Horizon-free reinforcement learning in adversarial linear mixture MDPs. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [8] Weng Lam Tam*, Xiao Liu*, **Kaixuan Ji**, Lilong Xue, Jiahua Liu, tao li, Yuxiao Dong, and Jie Tang. Parameter-efficient prompt tuning makes generalized and calibrated neural text retrievers. In *Findings of the Association for Computational Linguistics: EMNLP 2023 (EMNLP-Findings)*, 2023.
- [9] Xiao Liu*, **Kaixuan Ji***, Yicheng Fu*, Weng Lam Tam, Zhengxiao Du, Zhilin Yang, and Jie Tang. P-tuning: Prompt tuning can be comparable to fine-tuning across scales and tasks. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

MANUSCRIPTS UNDER REVIEW

- [1] Kaixuan Ji, Qingyue Zhao, Heyang Zhao, Qiwei Di, and Quanquan Gu. Near-optimal regret for KL-regularized multi-armed bandits. In *The 1st Workshop on Scaling Post-training for LLMs*, 2026.

ACADEMIC SERVICES

Conference and Journal Reviewer

- International Conference on Machine Learning (ICML): 2025, 2026
- Annual Conference on Neural Information Processing Systems (NeurIPS): 2024, 2025
- International Conference on Learning Representations (ICLR): 2024, 2025, 2026
- Association for the Advancement of Artificial Intelligence (AAAI): 2026
- International Conference on Artificial Intelligence and Statistics (AISTATS): 2024, 2025, 2026
- IEEE Transactions on Information Theory (TIT)

TEACHING

Teaching Assistant

Fall 2024, Winter 2025, Spring 2025, Fall 2025
CS35L Software Construction, University of California, Los Angeles